

ESPARGOS: Phase-Coherent WiFi CSI Datasets for Wireless Sensing Research

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Abstract—The use of WiFi signals to sense the physical environment is gaining popularity, with some common applications being motion detection and transmitter localization. Standard-compliant WiFi provides a cost effective, easy and backward-compatible approach to Joint Communication and Sensing and enables a seamless transfer of results from experiments to practical applications. However, most WiFi sensing research is conducted on channel state information (CSI) data from current-generation devices, which are usually not meant for sensing applications and thus lack sufficient spatial diversity or phase synchronization. With ESPARGOS, we previously developed a phase-coherent, real-time capable many-antenna WiFi channel sounder specifically for wireless sensing. We describe how we use ESPARGOS to capture large CSI datasets that we make publicly available. The datasets are extensively documented and labeled, for example with information from reference positioning systems, enabling data-driven and machine learning-based research.

Index Terms—WiFi Sensing, Channel State Information, Joint Communication and Sensing, Channel Charting, MIMO

I. INTRODUCTION

Neural networks and other machine learning technologies are widely regarded as key enablers for next-generation wireless systems and have long been a focal point in telecommunications research. In fields like computer vision and natural language processing, the availability of standardized training datasets has facilitated the development, performance evaluation, and comparison of machine learning algorithms. However, in wireless research, particularly within the context of massive multiple-input multiple-output (MIMO) systems, the availability of high-quality channel state information (CSI) datasets remains limited [1], forcing researchers to rely on channel models or simulated data. To address this gap, we introduced DICHASUS [2], a distributed massive MIMO channel sounder, which we regularly use to generate and publish highly calibrated CSI data¹. These datasets have been applied across various fields, including differentiable ray tracing [3], Channel Charting [4], channel modeling, and other work related to Joint Communication and Sensing (JCaS) [5]. Channel sounders like DICHASUS are designed specifically to collect high-quality CSI, and are typically neither real-time capable nor standards-compliant, often relying on software-defined radios (SDRs) at both ends of the link.

In a related line of research, it was observed that some commercially available WiFi chips offer CSI dumping functional-



Fig. 1. ESPARGOS antenna array with 2×4 patch antennas.

ity, allowing CSI to be extracted and processed [6] [7]. The use of WiFi has an obvious advantage: research outcomes can be easily transferred from theory to practical application. However, most compatible devices offer limited spatial diversity (few antennas), may lack meaningful synchronization, and, more broadly, were not designed with sensing applications in mind. To combine the spatial diversity of custom channel sounders with the flexibility and compatibility of commercial WiFi chip-based solutions, we developed ESPARGOS [8], a low-cost, WiFi-based channel sounder. Although ESPARGOS uses off-the-shelf WiFi chips, our circuit board design ensures phase synchronization across an arbitrary number of WiFi antennas. Both hardware and software were designed with sensing applications in mind, facilitating the development of real-time applications without the need to tinker with WiFi drivers to obtain CSI. Even though ESPARGOS is real-time capable, the benefits of dataset-driven research still apply, most notably the comparability of results and the quick and easy access to datasets measured for environments and scenarios of interest. For these reasons, we now also publish CSI datasets measured with ESPARGOS². In the following, we summarize key features of the ESPARGOS system (Sec. II), document the collection and processing of WiFi CSI datasets (Sec. III) and, using Channel Charting as an example, demonstrate how ESPARGOS datasets can accelerate the development of data-driven, real-time capable applications (Sec. IV). Finally, we conclude with an outlook that also draws a comparison to DICHASUS (Sec. V).

II. SUMMARY: ESPARGOS HARDWARE AND SOFTWARE

At its current stage of development, an ESPARGOS antenna array consists of eight antennas, each connected to an off-the-shelf WiFi-capable microcontroller (Espressif ESP32-S2), arranged into two rows of four antennas as shown in Fig. 1. Over a bus interface, measured CSI from all antennas is streamed to

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¹DICHASUS datasets are available at <https://dichasus.inue.uni-stuttgart.de/>

²ESPARGOS datasets are available at <https://espargos.net/>

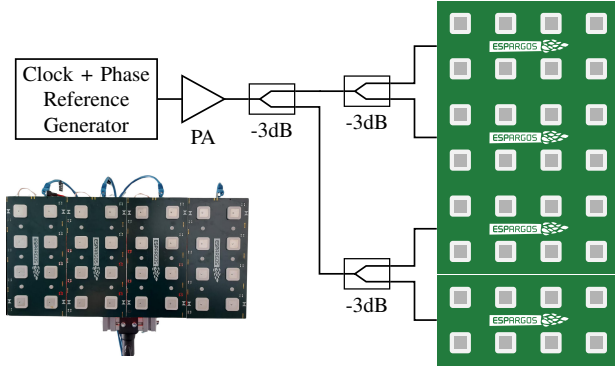


Fig. 2. With a common clock and phase reference signal, multiple ESPARGOS devices can be combined into a large phase-synchronous antenna array (schematic diagram and cropped photo of combined 4×8 array)

a central controller, which aggregates the measurements and forwards them over an Ethernet connection. As thoroughly explained in [8], by clocking all microcontrollers from the same crystal oscillator and by distributing a phase reference signal to compensate for phase ambiguities introduced by the phase locked loops (PLLs), the CSI measurements are synchronized in phase. ESPARGOS is usually operated as a passive sniffer, i.e., it listens for WiFi transmissions by other devices, but never transmits anything by itself. This way, it can be easily integrated into existing WiFi deployments. When collecting datasets for publication, however, we use a dedicated transmitter device. This device was specially configured to continuously transmit very short WiFi packets, which ensures that up-to-date CSI is available in regular intervals.

Multiple ESPARGOS boards can be combined into larger antenna arrays: To this end, the clock signal (at 40 MHz) and the phase reference signal (at around 2.4 – 2.5 GHz), which are both generated by the central controller, are frequency-multiplexed so that they can be carried over a single coaxial cable. The resulting combined reference signal is then amplified by a suitable Power Amplifier (PA) and distributed to all ESPARGOS boards over a cascade of power splitters, as shown in Fig. 2. If matching cable lengths are chosen and the power splitters offer sufficiently small phase unbalance, measured CSI from all ESPARGOS boards is phase-coherent. Otherwise, a constant phase offset may be introduced, which can, however, be accounted for in software. To control multiple ESPARGOS devices from a central computer, a Python library called *pyespargos*³ manages the board configuration and handles streaming CSI in real-time. A suite of demo applications shows how to use the library for applications like angle of arrival estimation. To provide measurement datasets, a special application collects CSI data provided by *pyespargos* as well as metadata and writes everything to a file.

III. DATASET COLLECTION AND PUBLICATION

CSI datasets are collections of CSI datapoints and associated metadata. In the case of ESPARGOS, CSI is estimated based on the high throughput long training field (HT-LTF) in each

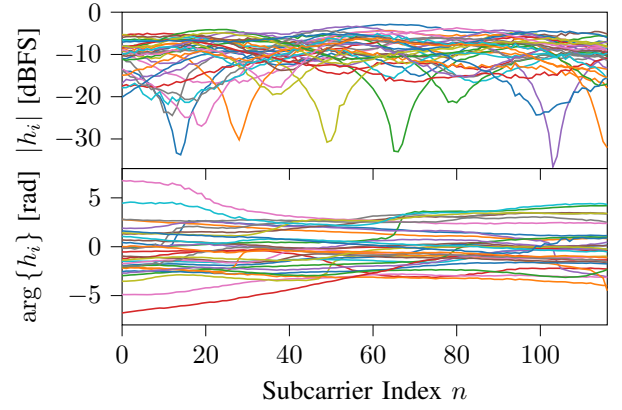


Fig. 3. Visualization of $\bar{\mathbf{H}} \in \mathbb{C}^{B \times 2 \times 4 \times N_{\text{sub}}}$ with $B = 4$ and $N_{\text{sub}} = 117$: Channel coefficient amplitude and phase over subcarrier index $n = 0, \dots, N_{\text{sub}} - 1$, each color representing one of the $B \times 2 \times 4$ antennas.

WiFi packet preamble. A suitable WiFi packet that is captured by all ESPARGOS receivers in the system hence constitutes a CSI datapoint, or, in other words, the channel coefficients contained within one CSI datapoint correspond to the channel estimates produced by all receivers for the same WiFi packet. The channel coefficients can be represented as a multidimensional array: Since the receivers provide channel coefficients for all N_{sub} subcarriers, a system with B ESPARGOS boards, each made up of 2 rows of 4 receivers, produces CSI datapoints that contain complex-valued frequency-domain channel coefficients $\mathbf{H} \in \mathbb{C}^{B \times 2 \times 4 \times N_{\text{sub}}}$. Exemplary frequency-domain CSI data $\bar{\mathbf{H}}$ is shown in Fig. 3. $\bar{\mathbf{H}}$ is the result of interpolating over 40 channel estimates $\mathbf{H}^{(l)}$ measured within an interval of 310 ms. The considered interval was arbitrarily selected from the *espargos-0002* [9] CSI dataset (see below).

In addition to channel coefficients, each receiver provides packet-specific received signal strength indicator (RSSI) values, which are collected in $\mathbf{P} \in \mathbb{R}^{B \times 2 \times 4}$. To account for the effect of variable receiver gains, channel coefficients may be weighted by the corresponding RSSI. When collecting CSI datasets with ESPARGOS, we make sure to also gather metadata such as the timestamp t (in seconds) and the exact location of the transmitter $\mathbf{x} \in \mathbb{R}^3$ at the time of transmission. To provide this data, which may be necessary “ground truth” for some machine learning applications, we use reference positioning systems such as a tachymeter total station (Leica MS60), which provides millimeter-level accuracy and high update rates. We also make sure to account for any time offsets in the system, making sure to align transmission timestamps to the timestamps of the reference positioning system. Combining CSI and metadata, a dataset $\mathcal{S}$ containing a total of L datapoints with indices l can be written as

$$\mathcal{S} = \{(\mathbf{H}^{(l)}, \mathbf{P}^{(l)}, \mathbf{x}^{(l)}, t^{(l)})\}_{l=1, \dots, L}.$$

In addition to datapoint-specific metadata, we also provide the exact location and orientation of all ESPARGOS arrays, photos of the environment and measurement setup and, for many datasets, even a 3-dimensional pointcloud scan of the environment, making sure that all relevant details of the

³Available at <https://github.com/ESPARGOS/pyespargos>

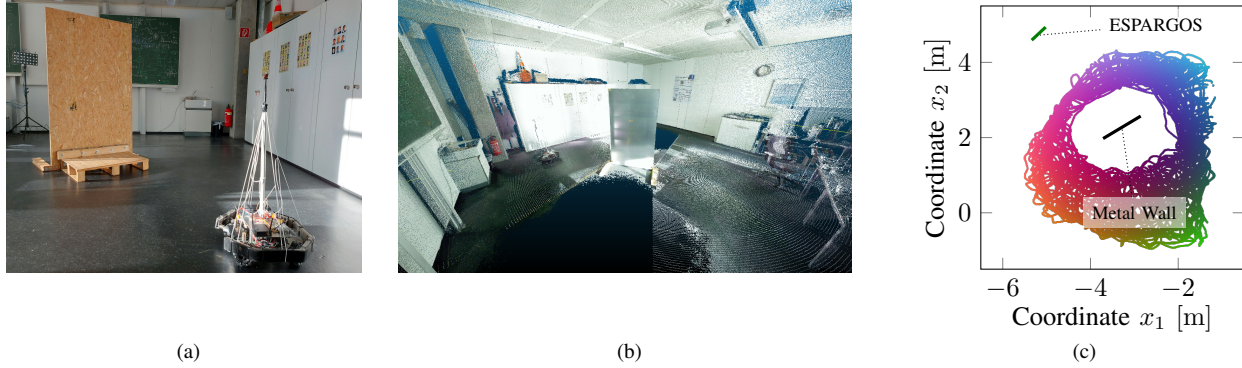


Fig. 4. The exemplary dataset *espargos-0002*: The figure shows (a) a photograph of the environment with the antenna array in the background, (b) a rendering of the 3D pointcloud and (c) a scatter plot (top view) of colorized “ground truth” positions of datapoints in $\mathcal{S}$, including antenna array and metal wall.

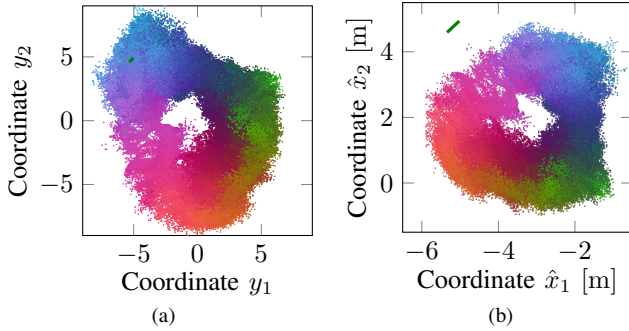


Fig. 5. Learned channel chart (a) before and (b) after optimal affine transformation, datapoint colorization is preserved from Fig. 4c

measurement setup are documented. All spatial information (antenna locations, pointcloud coordinates) is provided in the same right-handed cartesian coordinate system with meters as units, though the origin and orientation of the coordinate system are arbitrary. We use a publication process that assigns a unique Digital Object Identifier (DOI) to each dataset, which, apart from the unique name, ensures unambiguous citability.

In the following, we consider a dataset called *espargos-0002* [9], which is introduced in Fig. 4: As shown in Fig. 4a, the dataset was captured in an indoor lab room. As previously described, $B = 4$ ESPARGOS antenna arrays are combined into a single phase-synchronous 4×8 array, and the WiFi transmitter is mounted to a robot, which is tracked by a total station. By attaching the prism to the tip of the transmit antenna, which itself is on the tip of a pole, we ensure that the total station tracks the precise location of the antenna. The robot is programmed to follow customized trajectories in the measurement area. This way, as can be observed from Fig. 4c, datapoints were captured in the measurement area with a high spatial density. Due to the metal wall in the measurement area, which is drawn to scale in Fig. 4c and also visible in Fig. 4b, many locations in the measurement area experience mostly non-line of sight propagation.

IV. APPLICATION EXAMPLE: CHANNEL CHARTING

As one use case of ESPARGOS CSI datasets for research purposes, we consider the task of CSI-based indoor localization. From the earlier description of the *espargos-0002* dataset,

it is clear that there are no simple model-based approaches for indoor localization for the given scenario: While it is possible to determine an angle of arrival for datapoints with a line-of-sight (LoS) path between transmitter and receiver array, triangulation is not applicable with only one antenna array. Time of arrival-based multilateration is impossible because there is no time synchronization between transmitter and receiver. Due to the dominance of multipath propagation in the indoor environment, it is also not reasonable to use RSSI information to estimate the distance between transmitter and receiver based on path loss. Despite these difficult circumstances, a data-driven, self-supervised approach for indoor localization like Channel Charting [10] can at least provide relative position estimates without requiring reference positions. In contrast to model-based approaches like triangulation or multilateration, which assume a LoS channel, Channel Charting makes no such assumption about the propagation environment, but instead relies on similarity relationships between measured CSI samples. Since CSI is so high-dimensional, it is commonly assumed that every possible location of the transmitter in physical space manifests itself in a unique representation in CSI space, and that datapoints with transmitter locations which are close in physical space are also close in CSI space. Since the set of all possible locations of the transmitter is low-dimensional (in this case, two-dimensional) and the space of all possible channel realizations is high-dimensional, the problem can be understood as identifying and “un-folding” a low-dimensional manifold in high-dimensional CSI space. In that sense, Channel Charting is a dimensionality reduction techniques that finds a mapping $\mathcal{C}_\theta : \mathbb{C}^{B \times 2 \times 4 \times N_{\text{sub}}} \rightarrow \mathbb{R}^2$, called forward charting function (FCF), that maps high-dimensional CSI $\mathbf{H} \in \mathbb{C}^{B \times 2 \times 4 \times N_{\text{sub}}}$ to coordinates $\mathbf{y} \in \mathbb{R}^2$ in a low-dimensional space called the channel chart. This channel chart can be interpreted as a *map* of the environment and while it may be distorted compared to the physical reality, it should at least provide meaningful information about relative positions. Often, channel chart coordinates can be mapped to coordinates in physical space in a meaningful way by finding an affine transformation between these coordinate frames.

A detailed description of the Channel Charting algorithm is outside the scope of this work, but the Triplet Neural Network-based approach [11] taken here is similar to what

TABLE I
PERFORMANCE METRICS FOR CHANNEL CHARTING. METRICS MARKED WITH (*) WERE EVALUATED AFTER THE OPTIMAL AFFINE TRANSFORM T_c .

CT	TW	KS	MAE*	CEP*
0.96	0.96	0.20	0.44 m	0.42 m

CT = Continuity, TW = Trustworthiness, KS = Kruskal's Stress, MAE = Mean Abs. Err., CEP = Circular Error Probable, as defined in [13]

we describe in one of our previous publications [12], except for the different underlying dataset (now ESPARGOS and no longer DICHASUS) and tweaked hyperparameters⁴. For the following results, we use a subset of *espargos-0002*, containing $L = 569190$ datapoints, though good results are also achievable with significantly fewer datapoints. The resulting channel chart, shown in Fig. 5a, is obtained by applying the learned FCF to all datapoints in the dataset. To facilitate a quick visual inspection, the colors assigned to the datapoints based on their reference positions (see Fig. 4c) are preserved in the channel chart. If the color gradient is reproduced in the channel chart, as is the case here, this indicates that the FCF has managed to recover the global geometry of the environment. While some areas of the channel chart are somewhat distorted, the overall ring shape remains clearly visible. Obviously, the channel chart is rotated, scaled and also translated relative to the physical coordinate frame. We find the optimal affine coordinate transform $T_c(\mathbf{y}) = \hat{\mathbf{A}}\mathbf{y} + \hat{\mathbf{b}}$ from the channel chart's coordinate frame to physical coordinates by solving the least squares problem

$$(\hat{\mathbf{A}}, \hat{\mathbf{b}}) = \arg \min_{(\mathbf{A}, \mathbf{b})} \sum_{l=1}^L \|\mathbf{A}\mathbf{y}^{(l)} + \mathbf{b} - \mathbf{x}^{(l)}\|_2^2,$$

where $\mathbf{y}^{(l)} = \mathcal{C}_\theta(\mathbf{H}^{(l)})$ denotes the channel chart coordinates produced by the FCF for the datapoint with index l , and $\mathbf{x}^{(l)}$ is the corresponding two-dimensional "ground truth" position from the dataset. After T_c is applied to the channel chart, the resulting points $\hat{\mathbf{x}}^{(l)} = T_c(\mathcal{C}_\theta(\mathbf{H}^{(l)}))$ are now in the correct physical coordinate frame as shown in Fig. 5b.

Subjectively, the "ground truth" coordinates in Fig. 4c and the transformed channel chart in Fig. 5b look similar. A more objective performance assessment is conducted in Tab. I, which evaluates the channel chart based on performance metrics which are commonly used in Channel Charting literature. Compared to previous work on Channel Charting, the observed performance is acceptable, but leaves room for improvement.

V. CONCLUSION AND OUTLOOK

With ESPARGOS, we developed a real-time capable phase-coherent WiFi channel sounder. Owing to the benefits of dataset-driven research, we now also publish datasets containing CSI measured by ESPARGOS, alongside relevant metadata. Compared to data generated by our custom channel sounder DICHASUS (and most other SDR-based channel sounders), CSI measured by ESPARGOS is considerably more noisy and may exhibit additional impairments, as we lack

control over some aspects of the WiFi chip. The advantages of the WiFi-based approach are the much lower cost, backwards-compatibility and real-time capability, which partially makes up for the lower data quality. The successful training of a FCF for Channel Charting proves that the CSI quality is at least sufficient for this type of application.

Since ESPARGOS is designed to be easy-to-use, the possibility of third-party dataset contributions is opened up. This is in contrast to bespoke channel sounders, which usually require experienced operators. While datasets and some application examples are publicly accessible already, we plan to continue to publish additional software components over time. Applications developed with the help of measured datasets can be practically tested and demonstrated thanks to the real-time capability of ESPARGOS. For example, we developed a real-time demonstrator for Channel Charting internally, and we plan to publish more details on this demonstrator soon. Most importantly, we invite researchers to work with the publicly available ESPARGOS datasets.

REFERENCES

- [1] E. Björnson, L. Sanguinetti, H. Wymeersch, J. Hoydis, and T. L. Marzetta, "Massive MIMO is a reality—What is next?: Five promising research directions for antenna arrays," *Digital Signal Processing*, vol. 94, pp. 3–20, 2019.
- [2] F. Euchner, M. Gauger, S. Dörner, and S. ten Brink, "A Distributed Massive MIMO Channel Sounder for "Big CSI Data"-driven Machine Learning," in *25th ITG Workshop on Smart Antennas*, 2021.
- [3] J. Hoydis, F. Ait Aoudia, S. Cammerer, F. Euchner, M. Nimier-David, S. ten Brink, and A. Keller, "Learning Radio Environments by Differentiable Ray Tracing," *arXiv preprint*, Dec. 2023.
- [4] S. Taner, V. Palhares, and C. Studer, "Channel Charting in Real-World Coordinates with Distributed MIMO," *arXiv preprint arXiv:2406.13722*, 2024.
- [5] A. Foliadis, M. H. Castañeda, R. A. Stirling-Gallacher, and R. S. Thomä, "Transfer Learning for CSI-based Positioning with Multi-environment Meta-learning," *arXiv preprint arXiv:2405.11816*, 2024.
- [6] D. Halperin, W. Hu, A. Sheth, and D. Wetherall, "Tool Release: Gathering 802.11n Traces with Channel State Information," *ACM SIGCOMM CCR*, vol. 41, no. 1, p. 53, Jan. 2011.
- [7] Y. Xie, Z. Li, and M. Li, "Precise Power Delay Profiling with Commodity WiFi," in *Proceedings of the 21st Annual International Conference on Mobile Computing and Networking*, ser. MobiCom '15. New York, NY, USA: ACM, 2015, p. 53–64. [Online]. Available: <http://doi.acm.org/10.1145/2789168.2790124>
- [8] F. Euchner, T. Schneider, M. Gauger, and S. ten Brink, "ESPARGOS: An Ultra Low-Cost, Realtime-Capable Multi-Antenna WiFi Channel Sounder," in *26th International ITG Workshop on Smart Antennas*. VDE, 2023, pp. 1–6.
- [9] F. Euchner and S. ten Brink, "CSI Dataset espargos-0002: Larger combined antenna array, indoor lab room with metal wall, LoS and NLoS areas," 2024. [Online]. Available: <https://doi.org/doi:10.18419/darus-4456>
- [10] C. Studer, S. Medjkouh, E. Gönültaş, T. Goldstein, and O. Tirkkonen, "Channel Charting: Locating Users Within the Radio Environment Using Channel State Information," *IEEE Access*, 2018.
- [11] P. Ferrand, A. Decurninge, L. G. Ordoñez, and M. Guillaud, "Triplet-Based Wireless Channel Charting," in *GLOBECOM 2020-2020 IEEE Global Communications Conference*. IEEE, 2020, pp. 1–6.
- [12] F. Euchner, P. Stephan, M. Gauger, S. Dörner, and S. ten Brink, "Improving Triplet-Based Channel Charting on Distributed Massive MIMO Measurements," in *2022 IEEE 23rd International Workshop on Signal Processing Advances in Wireless Communication (SPAWC)*. IEEE, 2022, pp. 1–5.
- [13] F. Euchner, P. Stephan, and S. ten Brink, "Augmenting Channel Charting with Classical Wireless Source Localization Techniques," in *57th Asilomar Conference*, 2023.

⁴Source code for Channel Charting with ESPARGOS datasets is available at <https://github.com/Jeija/ESPARGOS-WiFi-ChannelCharting>